Experiment No 6

Network Diagnostic Commands: Tracert, Ping, Ifconfig and Netstat

Aim

To execute and observe network diagnostic commands and examine route tracing, host connectivity, network interface configuration, and active connections.

Algorithm

1. Open the command-line terminal or Command Prompt.

2. Execute `tracert www.saveetha.com` to trace the route to the destination host.

3. Execute `ping 172.18.64.1` to test connectivity to the specified host and observe packet statistics.

4. Execute `ifconfig` in Kali Linux to display network interface details. The supplied screenshot shows interface information for `eth0` and `lo`.

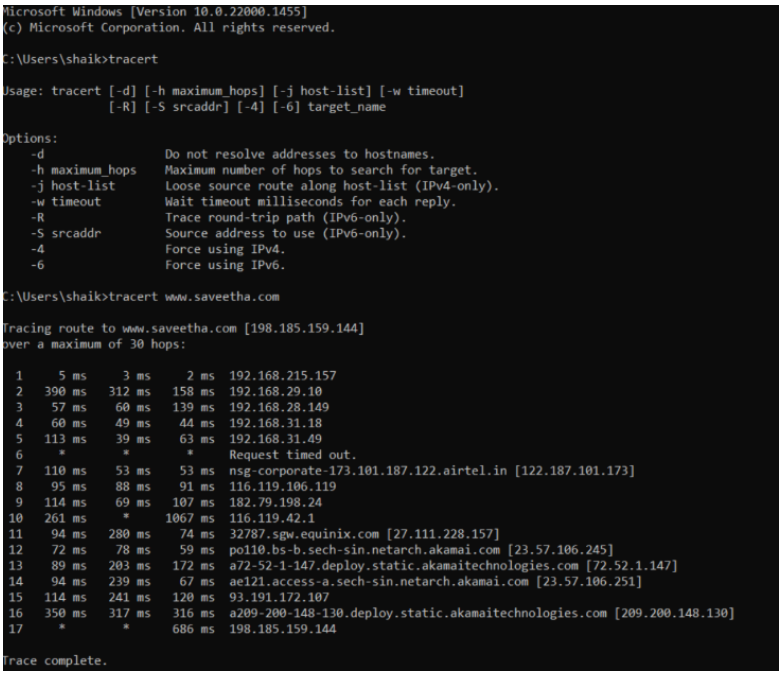
5. Execute `netstat` to display active network connections and their states.

6. Record the output and note any timeouts or connection details.

Output

The screenshots supplied for Experiment 6 are reproduced below.

Output Screenshot 1



```
Microsoft Windows [Version 10.0.22000.1455]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shaik>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                 Do not resolve addresses to hostnames.
    -h maximum_hops    Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                 Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                 Force using IPv4.
    -6                 Force using IPv6.

C:\Users\shaik>tracert www.saveetha.com

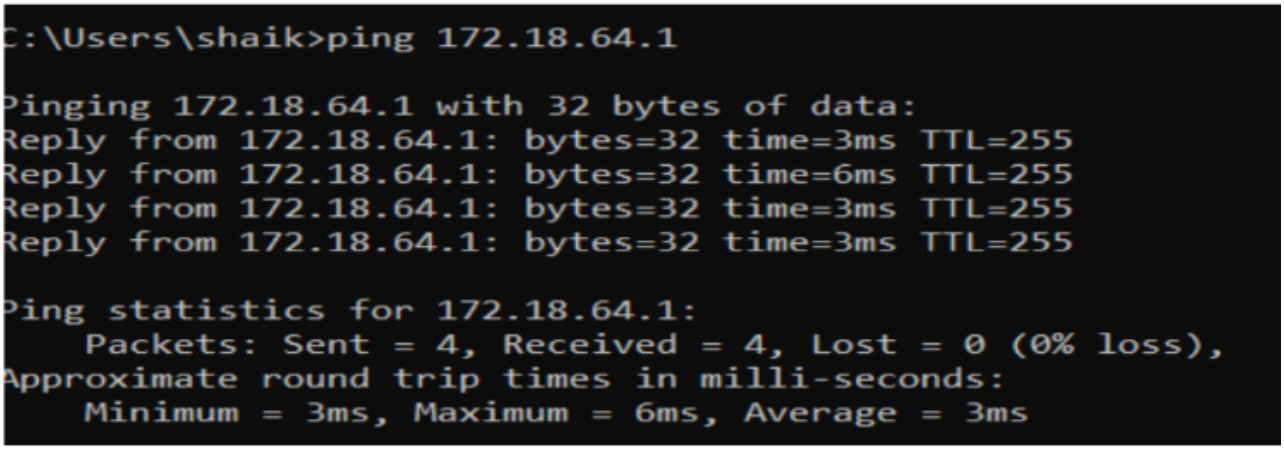
Tracing route to www.saveetha.com [198.185.159.144]
over a maximum of 30 hops:

  1    5 ms     3 ms     2 ms  192.168.215.157
  2   390 ms   312 ms   158 ms  192.168.29.10
  3    57 ms    60 ms   139 ms  192.168.28.149
  4    60 ms    49 ms    44 ms  192.168.31.18
  5   113 ms    39 ms    63 ms  192.168.31.49
  6     *        *        *     Request timed out.
  7   110 ms    53 ms    53 ms  nsg-corporate-173.101.187.122.airtel.in [122.187.101.173]
  8    95 ms    88 ms    91 ms  116.119.106.119
  9   114 ms    69 ms   107 ms  182.79.198.24
 10   261 ms     *     1067 ms  116.119.42.1
 11    94 ms   280 ms    74 ms  32787.sgw.equinix.com [27.111.228.157]
 12    72 ms    78 ms    59 ms  po110.bs-b.sech-sin.netarch.akamai.com [23.57.106.245]
 13    89 ms   203 ms   172 ms  a72-52-1-147.deploy.static.akamaitechnologies.com [72.52.1.147]
 14    94 ms   239 ms    67 ms  ae121.access-a.sech-sin.netarch.akamai.com [23.57.106.251]
 15   114 ms   241 ms   120 ms  93.191.172.107
 16   350 ms   317 ms   316 ms  a209-200-148-130.deploy.static.akamaitechnologies.com [209.200.148.130]
 17     *        *      686 ms  198.185.159.144

Trace complete.
```

Traceroute output for www.saveetha.com, showing the route hops and one request timeout.

Output Screenshot 2



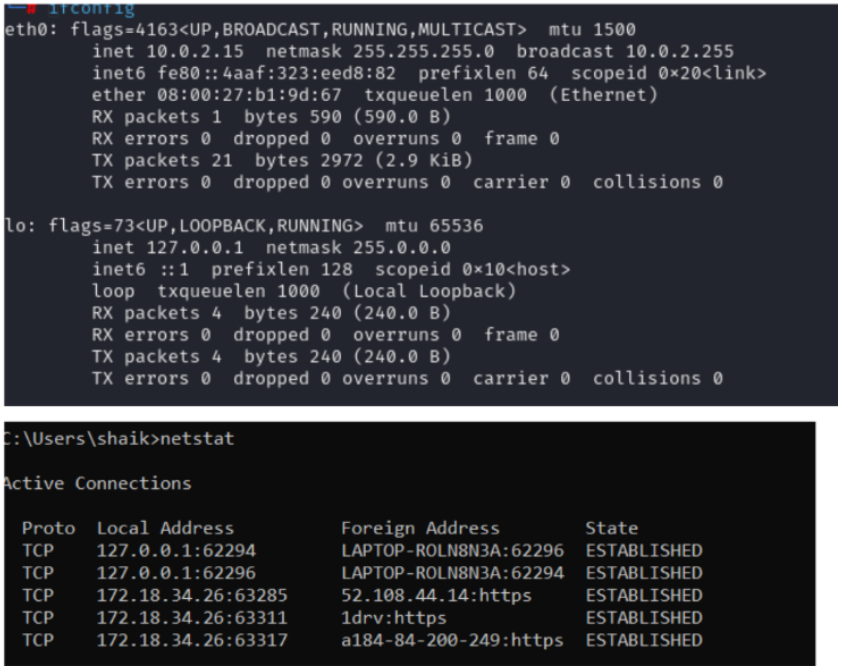
```
C:\Users\shaik>ping 172.18.64.1

Pinging 172.18.64.1 with 32 bytes of data:
Reply from 172.18.64.1: bytes=32 time=3ms TTL=255
Reply from 172.18.64.1: bytes=32 time=6ms TTL=255
Reply from 172.18.64.1: bytes=32 time=3ms TTL=255
Reply from 172.18.64.1: bytes=32 time=3ms TTL=255

Ping statistics for 172.18.64.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 6ms, Average = 3ms
```

Ping output for 172.18.64.1, showing four replies and 0% packet loss.

Output Screenshot 3



```
┌──# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::4aaf:323:eed8:82  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:b1:9d:67  txqueuelen 1000  (Ethernet)
        RX packets 1  bytes 590 (590.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 21  bytes 2972 (2.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 4  bytes 240 (240.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4  bytes 240 (240.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
C:\Users\shaik>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:62294        LAPTOP-ROLN8N3A:62296  ESTABLISHED
  TCP    127.0.0.1:62296        LAPTOP-ROLN8N3A:62294  ESTABLISHED
  TCP    172.18.34.26:63285     52.108.44.14:https     ESTABLISHED
  TCP    172.18.34.26:63311     1drv:https             ESTABLISHED
  TCP    172.18.34.26:63317     a184-84-200-249:https  ESTABLISHED
```

Network interface details from ifconfig and active connections from netstat.

Result

Thus, the network diagnostic commands were demonstrated. The supplied outputs show a traceroute to www.saveetha.com, successful ping replies with no packet loss for 172.18.64.1, interface configuration details, and active network connections.